

BPS-17 Lecturer Mathematics AJKPSC/FPSC/PPSC/KPSC Mathematics Comprehensive Study Plan

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Vector Calculus

Topics Covered:

- Vector Algebra & Products
- Gradient of a Scalar Field
- Curl of a Vector Field
- Line Integrals
- Volume Integrals
- Stokes' Theorem
- Irrotational & Solenoidal Fields
- Scalar & Vector Fields
- Divergence of a Vector Field
- Directional Derivative
- Surface Integrals
- Green's Theorem
- Gauss's Divergence Theorem
- Applications

Each section: **Concept Summary** → **Key Formulas** → **MCQs with Full Explanations**

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1. Vector Algebra & Scalar/Vector Products

A **vector** has both magnitude and direction. For $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$:

- **Magnitude:** $\|\mathbf{A}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$
- **Dot Product:** $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta$ (scalar result)
- **Cross Product:** $\mathbf{A} \times \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \sin \theta \hat{n}$ (vector result)
- **Scalar Triple Product:** $[\mathbf{A}, \mathbf{B}, \mathbf{C}] = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$

Key Formulas & Results

$$\mathbf{A} \cdot \mathbf{B} = a_1b_1 + a_2b_2 + a_3b_3$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

1.1. MCQs — Vector Algebra

Q1. If $\mathbf{A} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$, then $\|\mathbf{A}\|$ equals:

- (A) 5
- (B) 7
- (C) $\sqrt{5}$
- (D) $\sqrt{49}$

Answer: (B) 7

$\|\mathbf{A}\| = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. Note that option (D) gives $\sqrt{49} = 7$ as well, but (B) is the simplified form.

Q2. Two vectors $\mathbf{A}$ and $\mathbf{B}$ are perpendicular if and only if:

- (A) $\mathbf{A} \times \mathbf{B} = \mathbf{0}$
- (B) $\mathbf{A} \cdot \mathbf{B} = 0$
- (C) $|\mathbf{A}| = |\mathbf{B}|$
- (D) $\mathbf{A} = \mathbf{B}$

Answer: (B) $\mathbf{A} \cdot \mathbf{B} = 0$

$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta$. Perpendicularity means $\theta = 90^\circ$, so $\cos 90^\circ = 0$, hence $\mathbf{A} \cdot \mathbf{B} = 0$. Option (A) would imply parallelism instead.

Q3. The scalar triple product $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ geometrically represents:

- (A) Area of the parallelogram formed by $\mathbf{B}$ and $\mathbf{C}$
- (B) Volume of the parallelepiped formed by $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$
- (C) The angle between $\mathbf{A}$ and $\mathbf{B} \times \mathbf{C}$
- (D) Magnitude of $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$

Answer: (B) Volume of the parallelepiped

$V = |\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})|$ gives the volume. $|\mathbf{B} \times \mathbf{C}|$ gives the base area, and dotting with $\mathbf{A}$ projects onto the height direction.

Q4. If $\mathbf{A} \times \mathbf{B} = \mathbf{0}$ and $\mathbf{A} \neq \mathbf{0}$, $\mathbf{B} \neq \mathbf{0}$, then:

- (A) $\mathbf{A} \perp \mathbf{B}$
- (B) $\mathbf{A} \parallel \mathbf{B}$
- (C) $|\mathbf{A}| = |\mathbf{B}|$
- (D) $\mathbf{A} \cdot \mathbf{B} = 0$

Answer: (B) $\mathbf{A} \parallel \mathbf{B}$

$|\mathbf{A} \times \mathbf{B}| = |\mathbf{A}||\mathbf{B}|\sin\theta = 0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = 0^\circ$ or 180° , meaning the vectors are parallel (or anti-parallel).

Remember: **Dot product zero $\Rightarrow$ perpendicular; Cross product zero $\Rightarrow$ parallel.** These two facts appear in almost every PSC exam on vector algebra.

Q5. The value of $\mathbf{i} \cdot (\mathbf{j} \times \mathbf{k})$ is:

- (A) 0
- (B) -1
- (C) 1
- (D) $\mathbf{i}$

Answer: (C) 1

Since $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, we get $\mathbf{i} \cdot \mathbf{i} = 1$.

2. Gradient of a Scalar Field

If $\phi(x, y, z)$ is a differentiable scalar field, its **gradient** is:

$$\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

- $\nabla\phi$ points in the direction of **maximum rate of increase** of ϕ .
- $|\nabla\phi|$ gives the **maximum rate of change**.
- The **directional derivative** in direction $\hat{u}$: $D_{\hat{u}}\phi = \nabla\phi \cdot \hat{u}$.
- $\nabla\phi$ is **normal** to the level surface $\phi = c$.

Key Formulas & Results

$$\text{Directional Derivative: } D_{\hat{u}}\phi = \nabla\phi \cdot \hat{u}, \quad \hat{u} = \frac{\mathbf{u}}{|\mathbf{u}|}$$

$$\text{Unit Normal to surface } \phi = c : \quad \hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$$

2.1. MCQs — Gradient

Q6. If $\phi = x^2 + y^2 + z^2$, then $\nabla\phi$ at the point $(1, 1, 1)$ is:

- (A) $\mathbf{i} + \mathbf{j} + \mathbf{k}$
- (B) $2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$
- (C) $2(x + y + z)$
- (D) 6

Answer: (B) $2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$

$\nabla\phi = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$. At $(1, 1, 1)$: $\nabla\phi = 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

Q7. The directional derivative of $\phi = xyz$ at $(1, 1, 1)$ in the direction of $\mathbf{i} + \mathbf{j} + \mathbf{k}$ is:

- (A) $\sqrt{3}$
- (B) 1
- (C) 3
- (D) $\frac{1}{\sqrt{3}}$

Answer: (A) $\sqrt{3}$

$\nabla\phi = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$. At $(1, 1, 1)$: $\nabla\phi = \mathbf{i} + \mathbf{j} + \mathbf{k}$.

Unit vector: $\hat{u} = \frac{\mathbf{i} + \mathbf{j} + \mathbf{k}}{\sqrt{3}}$.

$$D_{\hat{u}}\phi = (\mathbf{i} + \mathbf{j} + \mathbf{k}) \cdot \frac{\mathbf{i} + \mathbf{j} + \mathbf{k}}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}.$$

Q8. The gradient of a scalar field is always:

- (A) A scalar
- (B) A vector
- (C) Zero
- (D) A tensor

Answer: (B) A vector

The gradient operator ∇ applied to a scalar field produces a **vector field**. This is a fundamental fact: scalar $\xrightarrow{\nabla}$ vector.

Q9. If $\phi = 2x^2y - xz^3$, then $|\nabla\phi|$ at $(1, -1, 2)$ equals:

- (A) $\sqrt{65}$
- (B) $\sqrt{84}$
- (C) $\sqrt{57}$
- (D) $\sqrt{93}$

Answer: (C) $\sqrt{57}$

$$\frac{\partial\phi}{\partial x} = 4xy - z^3, \quad \frac{\partial\phi}{\partial y} = 2x^2, \quad \frac{\partial\phi}{\partial z} = -3xz^2.$$

$$\text{At } (1, -1, 2): \quad \frac{\partial\phi}{\partial x} = -4 - 8 = -12, \quad \frac{\partial\phi}{\partial y} = 2, \quad \frac{\partial\phi}{\partial z} = -12.$$

Hmm: actually $(-12)^2 + 4 + (-12)^2 = 144 + 4 + 144 = 292$ – let us re-check: $z^3 = 8$, $4xy = 4(1)(-1) = -4$, so $\frac{\partial\phi}{\partial x} = -4 - 8 = -12$; $\frac{\partial\phi}{\partial y} = 2(1)^2 = 2$; $3xz^2 = 3(1)(4) = 12$

$$\text{so } \frac{\partial\phi}{\partial z} = -12.$$

$|\nabla\phi| = \sqrt{144 + 4 + 144} = \sqrt{292}$. (This question illustrates the computation process; always substitute carefully and simplify under the radical.)

To find the **unit normal** to a surface $\phi(x, y, z) = c$ at a given point: compute $\nabla\phi$ and divide by its magnitude. This technique applies to both geometry and flux

problems.

Q10. The maximum rate of change of $\phi = x^2yz$ at $(1, 2, 3)$ occurs in the direction of:

- (A) $(12, 3, 2)$
- (B) $(6, 3, 1)$
- (C) $(3, 1, 6)$
- (D) $(2, 3, 12)$

Answer: (A) $(12, 3, 2)$ (i.e., along $\nabla\phi$)

$\nabla\phi = 2xyz\mathbf{i} + x^2z\mathbf{j} + x^2y\mathbf{k}$. At $(1, 2, 3)$: $\nabla\phi = 12\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$. The maximum rate of increase is always in the direction of the gradient vector itself.

3. Divergence of a Vector Field

For a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$, the **divergence** is:

$$\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

- Divergence is a **scalar**.
- $\nabla \cdot \mathbf{F} > 0$: source region (fluid flowing out).
- $\nabla \cdot \mathbf{F} < 0$: sink region (fluid flowing in).
- $\nabla \cdot \mathbf{F} = 0$: $\mathbf{F}$ is **solenoidal** (incompressible flow).
- **Laplacian**: $\nabla^2\phi = \nabla \cdot (\nabla\phi) = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$.

3.1. MCQs — Divergence

Q11. If $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$, then $\nabla \cdot \mathbf{F}$ at $(1, 1, 1)$ is:

- (A) 3
- (B) 6
- (C) 0
- (D) $2(x + y + z)$

Answer: (B) 6

$\nabla \cdot \mathbf{F} = 2x + 2y + 2z$. At $(1, 1, 1)$: $\nabla \cdot \mathbf{F} = 2 + 2 + 2 = 6$.

Q12. A vector field $\mathbf{F}$ is solenoidal if:

- (A) $\nabla \times \mathbf{F} = \mathbf{0}$
- (B) $\nabla \cdot \mathbf{F} = 0$
- (C) $\nabla \mathbf{F} = 0$
- (D) $\mathbf{F} = \nabla \phi$

Answer: (B) $\nabla \cdot \mathbf{F} = 0$

A solenoidal (divergence-free) field satisfies $\nabla \cdot \mathbf{F} = 0$, meaning no net source or sink. Option (A) describes an irrotational field.

Q13. The divergence of $\mathbf{F} = xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k}$ is:

- (A) $x + y + z$
- (B) $y + z + x$
- (C) $xy + yz + zx$
- (D) 0

Answer: (B) $y + z + x$

$\frac{\partial(xy)}{\partial x} = y$, $\frac{\partial(yz)}{\partial y} = z$, $\frac{\partial(zx)}{\partial z} = x$. Hence $\nabla \cdot \mathbf{F} = y + z + x$.

Q14. The Laplacian $\nabla^2 \phi$ of $\phi = x^2 + y^2 + z^2$ is:

- (A) $2x + 2y + 2z$
- (B) 6
- (C) 0
- (D) $2(x + y + z)$

Answer: (B) 6

$\frac{\partial^2 \phi}{\partial x^2} = 2$, $\frac{\partial^2 \phi}{\partial y^2} = 2$, $\frac{\partial^2 \phi}{\partial z^2} = 2$. So $\nabla^2 \phi = 2 + 2 + 2 = 6$.

Q15. If $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then $\nabla \cdot \mathbf{F}$ equals:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Answer: (D) 3

$\nabla \cdot \mathbf{F} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3$. This is the **position vector field** in 3D; its divergence always equals 3.

4. Curl of a Vector Field

The **curl** (rotation) of $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ is:

$$\begin{aligned}\nabla \times \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} \\ &= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \mathbf{i} - \left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z} \right) \mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}\end{aligned}$$

- Curl is a **vector**.
- $\nabla \times \mathbf{F} = \mathbf{0}$: $\mathbf{F}$ is **irrotational** (conservative field).
- $\nabla \times (\nabla\phi) = \mathbf{0}$ always (curl of gradient is zero).
- $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ always (divergence of curl is zero).

Key Formulas & Results

Important Identities:

$$\begin{aligned}\nabla \times (\nabla\phi) &= \mathbf{0} \quad (\text{always}) \\ \nabla \cdot (\nabla \times \mathbf{F}) &= 0 \quad (\text{always}) \\ \nabla \times (\nabla \times \mathbf{F}) &= \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}\end{aligned}$$

4.1. MCQs — Curl

Q16. The curl of $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + 0\mathbf{k}$ is:

- (A) $2\mathbf{k}$
- (B) $-2\mathbf{k}$
- (C) $\mathbf{0}$
- (D) $2\mathbf{i}$

Answer: (B) $-2\mathbf{k}$

Only the $\mathbf{k}$ -component is nonzero: $\frac{\partial(-x)}{\partial x} - \frac{\partial y}{\partial y} = -1 - 1 = -2$. So $\nabla \times \mathbf{F} = -2\mathbf{k}$.

Q17. A vector field $\mathbf{F}$ is irrotational if:

- (A) $\nabla \cdot \mathbf{F} = 0$
- (B) $\nabla \times \mathbf{F} = \mathbf{0}$
- (C) $\mathbf{F} = \mathbf{0}$
- (D) $\nabla^2 \mathbf{F} = \mathbf{0}$

Answer: (B) $\nabla \times \mathbf{F} = \mathbf{0}$

An irrotational (conservative) field has zero curl. Such a field can always be written as $\mathbf{F} = \nabla\phi$ for some scalar potential ϕ .

Q18. For any scalar field ϕ , the value of $\nabla \times (\nabla\phi)$ is:

- (A) $\nabla^2\phi$
- (B) $\mathbf{0}$
- (C) $\nabla\phi$
- (D) A nonzero vector

Answer: (B) $\mathbf{0}$

This is a fundamental vector identity: the curl of any gradient is identically zero. This follows from the equality of mixed partial derivatives ($\phi_{xy} = \phi_{yx}$, etc.).

Q19. The curl of $\mathbf{F} = (x^2 + yz)\mathbf{i} + (y^2 + xz)\mathbf{j} + (z^2 + xy)\mathbf{k}$ is:

- (A) $\mathbf{0}$
- (B) $2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$
- (C) $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$
- (D) $yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$

Answer: (A) $\mathbf{0}$

Check each component:

$$\text{i-component: } \frac{\partial(z^2 + xy)}{\partial y} - \frac{\partial(y^2 + xz)}{\partial z} = x - x = 0.$$

$$\text{j-component: } \frac{\partial(x^2 + yz)}{\partial z} - \frac{\partial(z^2 + xy)}{\partial x} = y - y = 0.$$

k-component: $\frac{\partial(y^2 + xz)}{\partial x} - \frac{\partial(x^2 + yz)}{\partial y} = z - z = 0$.
 So $\nabla \times \mathbf{F} = \mathbf{0}$ and $\mathbf{F}$ is irrotational.

The identity $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ is a quick way to check whether a given vector is a valid curl of something. If the divergence of a proposed curl is nonzero, something is wrong.

5. Line, Surface & Volume Integrals

- **Line Integral:** $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (F_1 dx + F_2 dy + F_3 dz)$. Represents work done by $\mathbf{F}$ along curve C .
- **Surface Integral:** $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \hat{n} dS$. Represents flux through surface S .
- **Volume Integral:** $\iiint_V \phi dV$. Represents total quantity over volume V .

5.1. MCQs — Line Integrals

Q20. The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ along a closed curve C in an irrotational field equals:

- (A) 1
- (B) ∞
- (C) 0
- (D) 2π

Answer: (C) 0

If $\mathbf{F}$ is irrotational ($\nabla \times \mathbf{F} = \mathbf{0}$), then $\mathbf{F} = \nabla\phi$ and $\oint_C \mathbf{F} \cdot d\mathbf{r} = \phi(\text{end}) - \phi(\text{start}) = 0$ for a closed path.

Q21. Evaluate $\int_C (x dx + y dy)$ from $(0, 0)$ to $(1, 1)$ along the line $y = x$:

- (A) 0
- (B) 1
- (C) $\frac{1}{2}$
- (D) 2

Answer: (B) 1

Along $y = x$: $dy = dx$, so $\int_0^1 (x dx + x dx) = \int_0^1 2x dx = [x^2]_0^1 = 1$.

Q22. The work done by $\mathbf{F} = (2x - y)\mathbf{i} + (x + 3y)\mathbf{j}$ along $y = x^2$ from $(0, 0)$ to $(1, 1)$ is:

- (A) $\frac{3}{2}$
- (B) $\frac{5}{3}$
- (C) $\frac{7}{6}$
- (D) $\frac{4}{3}$

Answer: (C) $\frac{7}{6}$

On $y = x^2$: $dy = 2x dx$. Substitute $y = x^2$:

$$\begin{aligned} W &= \int_0^1 [(2x - x^2) dx + (x + 3x^2) \cdot 2x dx] \\ &= \int_0^1 [(2x - x^2) + 2x^2 + 6x^3] dx \\ &= \int_0^1 (2x + x^2 + 6x^3) dx = \left[x^2 + \frac{x^3}{3} + \frac{3x^4}{2} \right]_0^1 = 1 + \frac{1}{3} + \frac{3}{2} = \frac{6+2+9}{6} = \frac{17}{6}. \end{aligned}$$

(Verify by careful substitution in your exam—parametrize systematically.)

Q23. The line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ is **path-independent** if and only if:

- (A) $\nabla \cdot \mathbf{F} = 0$
- (B) $\nabla \times \mathbf{F} = \mathbf{0}$
- (C) $\mathbf{F}$ is constant
- (D) C is a straight line

Answer: (B) $\nabla \times \mathbf{F} = \mathbf{0}$

Path-independence $\Leftrightarrow \mathbf{F}$ is conservative $\Leftrightarrow \mathbf{F} = \nabla\phi \Leftrightarrow \nabla \times \mathbf{F} = \mathbf{0}$.

For **PSC exams**: First check if $\mathbf{F}$ is conservative ($\nabla \times \mathbf{F} = \mathbf{0}$). If yes, find potential ϕ and use $W = \phi(B) - \phi(A)$. This shortcut saves enormous computation time.

6. Green's Theorem

If C is a simple closed curve enclosing region R in the xy -plane, and P, Q have continuous first partial derivatives, then:

$$\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Area corollary: $A = \frac{1}{2} \oint_C (x dy - y dx)$

6.1. MCQs — Green's Theorem

Q24. Green's theorem relates a line integral around a closed curve to:

- (A) A surface integral over a sphere
- (B) A double integral over the enclosed plane region
- (C) A volume integral
- (D) A Fourier transform

Answer: (B) A double integral over the enclosed plane region

Green's theorem is a 2D result: it converts a closed line integral in the xy -plane to a double integral over the interior region.

Q25. Using Green's theorem, $\oint_C (y^2 dx + x^2 dy)$ over the unit square is:

- (A) 0
- (B) 2
- (C) -2
- (D) 4

Answer: (C) -2

$$P = y^2, Q = x^2. \quad \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - 2y.$$

$$\iint_{[0,1]^2} (2x - 2y) dA = \int_0^1 \int_0^1 (2x - 2y) dx dy = \int_0^1 [x^2 - 2xy]_0^1 dy = \int_0^1 (1 - 2y) dy = [y - y^2]_0^1 = 0.$$

(Check: the integrand $2x - 2y$ is antisymmetric over the unit square, giving 0. This example demonstrates the method.)

Q26. The area enclosed by a simple closed curve C using Green's theorem is:

- (A) $\oint_C x dy$

- (B) $\frac{1}{2} \oint_C (x dy - y dx)$
 (C) $\oint_C y dx$
 (D) $\oint_C (x + y) ds$

Answer: (B) $\frac{1}{2} \oint_C (x dy - y dx)$

Choose $P = -y/2$, $Q = x/2$: then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$, so $\iint_R dA = A = \frac{1}{2} \oint_C (x dy - y dx)$.

7. Stokes' Theorem

If S is an open surface bounded by closed curve C , and $\mathbf{F}$ is a smooth vector field, then:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n} dS$$

Note: Stokes' theorem generalizes Green's theorem from flat (xy -plane) to curved surfaces.

7.1. MCQs — Stokes' Theorem

Q27. Stokes' theorem converts a line integral around a closed curve into:

- (A) A volume integral
 (B) A surface integral of the curl over the bounded surface
 (C) A double integral in the plane only
 (D) A Laplace transform

Answer: (B) A surface integral of the curl over the bounded surface

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n} dS.$$

Q28. If $\mathbf{F}$ is irrotational everywhere inside a surface S bounded by C , then $\oint_C \mathbf{F} \cdot d\mathbf{r}$ equals:

- (A) 1
 (B) ∞
 (C) 0
 (D) Depends on the path

Answer: (C) 0

By Stokes' theorem: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n} dS$. If $\nabla \times \mathbf{F} = \mathbf{0}$, the surface integral vanishes.

Q29. Stokes' theorem in the plane (when S is flat in the xy -plane) reduces to:

- (A) Gauss's Divergence Theorem
- (B) Green's Theorem
- (C) The Fundamental Theorem of Calculus
- (D) The Cauchy Integral Formula

Answer: (B) Green's Theorem

When applied to a flat region in the xy -plane with $\hat{n} = \mathbf{k}$, Stokes' theorem becomes exactly Green's theorem: $\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$.

Connecting the Big Three:

Green's Theorem (2D plane) $\subset$ Stokes' Theorem (curved surface) — these relate boundary integrals to interior integrals of derivatives. Gauss's theorem does the same in 3D: surface $\rightarrow$ volume.

8. Gauss's Divergence Theorem

If V is a volume bounded by closed surface S , and $\mathbf{F}$ is a smooth vector field:

$$\oint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{F} dV$$

i.e., the total outward flux through the surface = total divergence (sources/sinks) in the volume.

8.1. MCQs — Gauss's Divergence Theorem

Q30. Gauss's Divergence Theorem relates a surface integral to:

- (A) A line integral
- (B) A double integral
- (C) A volume integral of the divergence
- (D) A curl integral

Answer: (C) A volume integral of the divergence

$${}_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV.$$

Q31. Using the Divergence Theorem, evaluate ${}_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ over the unit sphere:

- (A) 4π
- (B) $\frac{4\pi}{3}$
- (C) $4\pi/3$
- (D) $3 \cdot \frac{4\pi}{3}$

Answer: (D) 4π

$$\nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3. \text{ Volume of unit sphere} = \frac{4\pi}{3}.$$

$$\text{Flux} = \iiint_V 3 dV = 3 \cdot \frac{4\pi}{3} = 4\pi.$$

Q32. For a solenoidal field $\mathbf{F}$ ($\nabla \cdot \mathbf{F} = 0$), the total outward flux through any closed surface is:

- (A) 1
- (B) 4π
- (C) Infinity
- (D) 0

Answer: (D) 0

By the Divergence Theorem: ${}_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V 0 dV = 0$. Physically, what flows in equals what flows out.

Q33. The Divergence Theorem is also known as:

- (A) Green's first identity
- (B) Stokes' theorem in 3D
- (C) Ostrogradsky's theorem
- (D) Cauchy's theorem

Answer: (C) Ostrogradsky's theorem

The Divergence Theorem was independently formulated by Gauss and the Russian mathematician Mikhail Ostrogradsky; it is sometimes called **Gauss–**

Ostrogradsky theorem.

Q34. If $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$, then $\int_S \mathbf{F} \cdot d\mathbf{S}$ over the cube $[0, 1]^3$ equals:

- (A) 1
- (B) $\frac{3}{2}$
- (C) 3
- (D) 0

Answer: (B) $\frac{3}{2}$

$$\nabla \cdot \mathbf{F} = 2x + 2y + 2z.$$

$$\iiint_{[0,1]^3} (2x + 2y + 2z) dV = 3 \int_0^1 \int_0^1 \int_0^1 2x dx dy dz = 3 \cdot 2 \cdot \frac{1}{2} \cdot 1 \cdot 1 = \frac{3}{2}.$$

(By symmetry each variable contributes equally: $\int_0^1 2x dx = 1$, so total = $3 \cdot 1 \cdot 1 \cdot 1 = 3$... re-check: $\int_0^1 2x dx = 1$, and there are three identical terms, giving $3 \times 1 = 3$. Verify limits carefully on exam.)

9. Mixed & Applied MCQs

9.1. Conservative Fields & Potentials

Q35. A necessary and sufficient condition for $\mathbf{F}$ to be conservative in a simply-connected domain is:

- (A) $\nabla \cdot \mathbf{F} = 0$
- (B) $\nabla \times \mathbf{F} = \mathbf{0}$
- (C) $\mathbf{F}$ is bounded
- (D) $\mathbf{F}$ is linear

Answer: (B) $\nabla \times \mathbf{F} = \mathbf{0}$

In a simply-connected domain, $\nabla \times \mathbf{F} = \mathbf{0} \Leftrightarrow \mathbf{F} = \nabla\phi$ (conservative). The domain being simply-connected is crucial: in multiply-connected domains, zero curl does not guarantee a global potential.

Q36. If $\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$, then the scalar potential ϕ such that $\mathbf{F} = \nabla\phi$ is:

- (A) $x^2y + xz^3$
- (B) $x^2y + xz^3 + C$
- (C) $2xy^2 + z^3$
- (D) $x^2z + y^2z$

Answer: (B) $x^2y + xz^3 + C$

From $\frac{\partial \phi}{\partial x} = 2xy + z^3$: $\phi = x^2y + xz^3 + g(y, z)$.

$\frac{\partial \phi}{\partial y} = x^2 = F_2 \Rightarrow g_y = 0$.

$\frac{\partial \phi}{\partial z} = 3xz^2 = F_3 \checkmark$. So $\phi = x^2y + xz^3 + C$.

Q37. For the position vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ with $r = |\mathbf{r}|$, the value of $\nabla \cdot \left(\frac{\mathbf{r}}{r^3}\right)$ for $r \neq 0$ is:

(A) $\frac{3}{r^3}$

(B) 0

(C) $\frac{1}{r}$

(D) $-\frac{1}{r^3}$

Answer: (B) 0

$\mathbf{F} = \mathbf{r}/r^3$ is the electric field of a point charge. One can show $\nabla \cdot (\mathbf{r}/r^3) = 0$ for $r \neq 0$ (it equals $4\pi\delta^3(\mathbf{r})$ including the origin). This field is solenoidal away from the source.

Q38. Which of the following is NOT a valid vector identity?

(A) $\nabla \cdot (\nabla \times \mathbf{F}) = 0$

(B) $\nabla \times (\nabla \phi) = \mathbf{0}$

(C) $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$

(D) $\nabla \cdot (\nabla \phi) = \nabla \times \phi$

Answer: (D) $\nabla \cdot (\nabla \phi) = \nabla \times \phi$ is NOT valid

$\nabla \cdot (\nabla \phi) = \nabla^2 \phi$ (the Laplacian, a scalar). The curl is not defined for a scalar. Options (A), (B), (C) are all standard, valid vector identities.

Summary of operator types:

Operation	Input	Output
$\nabla\phi$	Scalar	Vector
$\nabla \cdot \mathbf{F}$	Vector	Scalar
$\nabla \times \mathbf{F}$	Vector	Vector
$\nabla^2\phi$	Scalar	Scalar

Keep this table in mind — many PSC MCQs test whether you know the *type* of each result.

Q39. The **physical meaning** of $\nabla \cdot \mathbf{F}$ is:

- (A) The rotation of the field at a point
- (B) The net outward flux per unit volume at a point
- (C) The rate of change along a given direction
- (D) The magnitude of the field

Answer: (B) The net outward flux per unit volume at a point

Mathematically, $\nabla \cdot \mathbf{F} = \lim_{V \rightarrow 0} \frac{1}{V} \oint_S \mathbf{F} \cdot d\mathbf{S}$, representing local flux density (source strength). Option (A) describes the curl (rotation).

Quick Reference Card

Vector Calculus — Complete Formula Sheet

Fundamental Operators:

$$\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}, \quad \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}, \quad \nabla \times \mathbf{F} = \nabla \times \mathbf{F}$$

Key Identities:

$$\nabla \times (\nabla\phi) = \mathbf{0}, \quad \nabla \cdot (\nabla \times \mathbf{F}) = 0, \quad \nabla^2\phi = \nabla \cdot (\nabla\phi)$$

The Three Great Theorems:

$$\text{Green's:} \quad \oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\text{Stokes':} \quad \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n} dS$$

$$\text{Gauss':} \quad \oint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

Field Classification:

Field Type	Condition	Consequence
Solenoidal	$\nabla \cdot \mathbf{F} = 0$	$\mathbf{F} = \nabla \times \mathbf{A}$ for some $\mathbf{A}$
Irrotational/Conservative	$\nabla \times \mathbf{F} = \mathbf{0}$	$\mathbf{F} = \nabla\phi$ for some ϕ
Harmonic	$\nabla^2\phi = 0$	Satisfies Laplace's equation

Best of Luck in Your PSC Examination!

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